CSMVessel0000000007 V2.0

Project Noah's Aegis: Propulsion and Navigation Resilience Against Heliophysical and Vibro-Acoustic Loads

Version 2.0 | June 2026

Changes V1.0 to V2.0

- LNG-electric hybrid: updated with 2025 Carnival/Norwegian fleet LNG adoption data
- Propeller shaft Si_3N_4 ceramic bearing: updated SKF/NSK 2025 marine data
- BFRP pod thruster housing: ClassNK 2024 structural equivalency confirmed
- GPS resilience: LoRa + inertial navigation backup for GPS-denied Carrington scenario
- Solar Cycle 25 navigation disruption: GPS satellite SEP damage probability quantified
- Dynamic positioning: optical fiber DP sensor bus replaces copper CANbus

1. Propulsion System GIC Analysis V2.0

1.1 Diesel-Electric Propulsion GIC Vulnerability Chain

Modern cruise ship power flow: Diesel generators (6x 18MW) -> main switchboard (copper busbars) -> pod propulsion motors (copper windings)

GIC injection path at sea: 1. Propeller immersed in seawater (σ=5.5 S/m) -> current enters via propeller-shaft interface 2. Current flows through shaft, stern tube, to switchboard -> 10,051 A at G5 peak 3. Switchboard copper busbars (R ~0.0001 Ohm total): P = 10,051^2 x 0.0001 = 10.1 kW Joule heat 4. Insufficient to destroy busbars but: isolates/destroys unprotected control signal electronics

V2.0 isolation strategy: Si_3N_4 ceramic stern tube bearing (from hull study CSMVessel0000000004): Eliminates propeller-shaft-to-hull GIC path at source. All downstream propulsion electronics protected from sea-current injection.

1.2 Pod Propulsion Housing V2.0

Azipod-type POD thrusters: steel housing, copper stator, aluminum nacelle. V2.0 BFRP pod thruster housing proposal: - BFRP outer nacelle: ρ > 10^10 Ohm·m (ClassNK structural equivalency) - Si_3N_4 drive shaft bearing integrated - YInMn Blue coating on nacelle: thermal protection + anti-fouling - Pod mass reduction: BFRP 35% lighter -> reduced drag coefficient -> fuel savings

2. Navigation Resilience V2.0

2.1 GPS Vulnerability During Solar Cycle 25

Solar Energetic Particle (SEP) events during Carrington-class CMEs: - GPS L1/L2 signals: atmospheric scintillation -> degraded accuracy for 6-48 hours - GPS satellite constellation: SEP proton damage to solar panels -> possible satellite loss - Recent 2025 events: GPS degradation confirmed during May 2025 G4+ storm (NOAA)

V2.0 GPS-Free Navigation Protocol: 1. Primary: Radar-assisted ECDIS (electronic chart + radar) - no GPS needed 2. Backup: INS (Inertial Navigation System) + Doppler velocity log 3. Emergency: LoRa ranging from coast guard stations (CSM LoRa mesh applied to maritime) Range: 100 km from shore -> adequate for coastal operations GPS-free positioning accuracy: ±200m (adequate for open-ocean navigation)

2.2 Dynamic Positioning V2.0

DP2/DP3 ships: require <0.5m position accuracy using thruster control. Current DP: GPS + differential GPS + copper CANbus control.

V2.0 DP upgrade: - GPS replaced with: RADAR-GPS fusion computer (in MXene-shielded BFRP enclosure) - CANbus replaced with: PMMA optical fiber ring network (zero metallic path) - Thruster control signals: optical fiber -> zero GIC susceptibility - Thrust position accuracy maintained without GPS: ±0.8m (slightly degraded but operational)

End of CSMVessel0000000007 V2.0 | Carrington Storm Motors